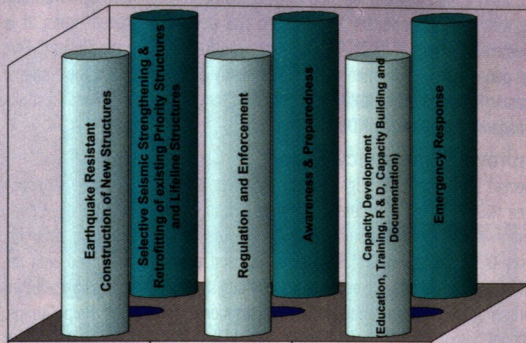


- Regulation and Enforcement
- Awareness and Preparedness
- Capacity Development (Education, Training, R & D, Capacity Building and Documentation)
- Emergency Response

The six pillars of earthquake management in India are shown below.



It is necessary to ensure that all new structures which are being built in the country henceforth will be built in strict compliance to the National Building Codes and other relevant safety codes so that even if a high intensity earthquake happens, the newly constructed assets and infrastructure will not collapse and cause loss of lives and damage to property, assets and infrastructure. Simultaneously, special efforts will have to be taken to undertake the structural safety audit of critical infrastructure and lifeline assets and wherever necessary, weak structures will have to be seismically strengthened and retrofitted. Given the large number of existing structures and building stock in the country estimated to be more than 250 million structures, it will not be practical or realistic to undertake the strengthening of such a large number of existing structures. However, critical infrastructure and lifeline assets will have to be evaluated for their capacity to withstand high intensity earthquakes, especially if they are in the high or very high earthquake-prone areas of seismic zones IV and V.

The rigorous enforcement of a techno-legal regime and strict compliance to existing earthquake-resistant building codes and other relevant safety codes has to be continuously monitored, especially in the high and very high risk seismic zones IV and V. The Model Town Planning Bye-law circulated by the

Government of India to the state governments in 2004 has to be reviewed and adopted to ensure the safety, especially in high risk areas. The enforcement personnel in Urban Local Bodies and other local administration agencies will have to familiarize themselves with the provisions of the relevant techno-legal regime and ensure compliance when evaluating building plans and issuing approvals and building permissions.

All stakeholders have to be aware about the earthquake risk and vulnerability in their local areas and strengthen the multi-stakeholder preparedness to improve local capacity to withstand the devastating impact of future earthquakes and if earthquakes strike, to have adequate capacity to respond effectively. State Governments have been encouraged to set up State Disaster Response Forces from their existing strength of police personnel who can be trained and equipped to respond effectively to natural disasters and man-made disasters.

Special Efforts in Earthquake Risk Mitigation

NDMA has initiated the steps for launching a National Earthquake Risk Mitigation Project (NERMP) with the help of the concerned Ministries of the Government of India and state governments to improve earthquake management in the country. Some of the major activities proposed to be carried out are listed below.

It is estimated that more than 800,000 engineers, over 25,000 architects and over one crore masons are associated with the construction industry in India. However, most of them are not adequately equipped with the knowledge and skills associated with the earthquake-resistant construction and seismic strengthening and retrofitting of weak structures. In 2004, the Government of India had initiated two programmes called National Programme for Capacity Building of Engineers in Earthquake Risk Management (NPCBEERM) and National Programme for Capacity Building of Architects in Earthquake Risk Management (NPCBAERM) with the help of reputed engineering education institutions like Indian Institutes of Technology and National Institutes of Technology. It is proposed to further intensify this programme by extending it through a larger number of institutions including selected Engineering Colleges as well as Industrial Training Institutes and Polytechnics for training engineers and masons respectively. Thus, the capacity building of engineers, architects, lead masons and masons in earthquake-resistant construction, structural safety audit and retrofitting will be carried out through a network of resource institutions in the country like Indian Institutes of Technology, National Institutes of Technology, selected engineering colleges and architecture colleges, Industrial Training Institutes and Polytechnics, apart from other professional agencies. The possibility of using distance education through the internet and television channels will also be explored.

There is an urgent need for creating greater Public Awareness on earthquake risk, vulnerability and emergency response in the event of occurrence of any earthquake among the general public in earthquake-prone areas. It is envisaged that these activities will be carried out with the help of the state governments, district administrations and the Disaster Management Authorities at the state and district levels through the electronic and print media as well as through street plays, mobile multi-media vans, hoardings, wall paintings etc.

The consolidation and strengthening of the Techno-Legal Regime and the strict compliance of existing earthquake-resistant building codes and other relevant safety codes as well as updated Town Planning Bye-laws is a sine qua non for the effective management of earthquakes in India. With the help of the state governments and concerned Ministries of the Government of India, special efforts will be made to strengthen the techno-legal regime.

In all the major earthquakes in the country in the past, it has been observed that the health facilities like district hospitals, primary health centres, etc. faced damage and became non-functional. The uninterrupted functioning of health facilities after the occurrence of an earthquake has to be ensured in earthquake-prone areas which are likely to face high and very high intensity earthquakes. Special efforts will be made to undertake structural safety audit of a few district hospitals in high and very high risk zones and carry out seismic strengthening and retrofitting of the critical lifeline blocks in these district hospitals and improve their capacity to face high intensity earthquakes.

In India, it is observed that very little allocations are made for the institutional strengthening of academic, professional and research institutions to carry out

